



Strategic Insights from Technical to Management Level: Private Hospital Finance

A WORK BY:

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FOR THE CLASS ITC6004B1 - DATA VISUALIZATION - WINTER TERM 2026
FINAL PROJECT

PROJECT ORGANIZATION

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Who are we ?

A group of three hospitals in New York State

- **Maimonides:** Large Establishment in NYC
- **Hudson Valley:** Medium Establishment
- **St. Joseph's:** Regional smaller establishment

We replicate real-life financial scenarios and demonstrate how they can be communicated to stakeholders across different hierarchy levels: Technical, Director, and C-Level.

C-Level Dashboard to the CEO of the Group

Director Dashboard to the CFO

Technical Dashboard to the Operations & Analysts Staff

Understanding the Dataset: Data Preprocessing & EDA

Bottom - Up Approach

The Data Foundation

The Core Source: SPARCS NYS

Origin	New York State Statewide Planning and Research Cooperative System (SPARCS).
Description	De-identified inpatient discharge records.
Volume	100000+ unique patient encounters
Features	33 Features Clinical & Financial
Years	2018 & 2019 Last The Last Stable Baseline before Covid-19 anomalies

Clinical Features

Facility Name	The name of the hospital providing health services
Age Group	Age in years at time of discharge. Grouped into bins (0 to 17, 18 to 29, 30 to 49, 50 to 69, and 70 or Older)
Length of Stay (LOS)	The total number of patient days per row of the dataset
Severity of Illness	Minor (1), Moderate (2), Major (3), Extreme (4)
Admission Type	Elective, Emergency, Newborn, Trauma, Urgent
Diagnosis	Illness categorization for invoicing
ED Indicator	If Emergency Department was invoiced
Payment Type	Medicare Federal, ages > 65, younger people with certain disabilities Medicaid Joint Federal and State program for people with limited income and resources Private Health Insurance Coverage provided by commercial entities (like Blue Cross, Aetna, or UnitedHealthcare) often through an employer Self-Pay / Other Includes patients who pay out-of-pocket

Financial

Costs	Estimated Real cost of a discharge. Internal to the hospital
Charges	Total amount charged to the patient

Engineered Features

Charge-to-Cost Ratio (CCR)	Charges / Costs
Markup (Profit)	Charges - Costs
Markup % Profit-to-Cost	$(\text{Charges} - \text{Costs}) / \text{Costs} \times 100$
Costs per Patient per Day (CPD)	Costs / Patients x LOS
Revenue (Charges) per Patient per Day (RPD)	Charges / Patients x LOS
CPD to RPD	$\text{COST} / \text{Charges}$ (per Patient per Day)
Cross Subsidy Ratio (CSR)	$\text{Profit_CashCows} / \text{Profit_LossMakers}$
Categorical Re-mapping	Severity of Illness - Ranking Age - Ranking



The Raw Data Reality

Data Cleaning and Univariate analysis

Explored and Corrected Data types

Missing Values or NaNs (0.003%)

Columns with Low Variance

Duplicates

Outliers - Numerical Variables

Categorical Variables

	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max
Length of Stay	107682.0	5.055980	7.440349	1.00	1.0000	1.0000	2.00	3.000	5.0000	15.000	35.0000	120.00
Year	107682.0	2018.495682	0.499984	2018.00	2018.0000	2018.0000	2018.00	2018.000	2019.0000	2019.000	2019.0000	2019.00
Total Charges	107682.0	52578.678601	88236.674008	1500.00	4455.2673	6052.0000	17181.89	29873.045	55216.6475	168937.794	381260.6335	4409949.53
Total Costs	107682.0	15946.164155	22726.076002	88.27	791.1400	947.5245	5410.03	9974.455	18189.6025	49996.698	103110.3964	956351.23

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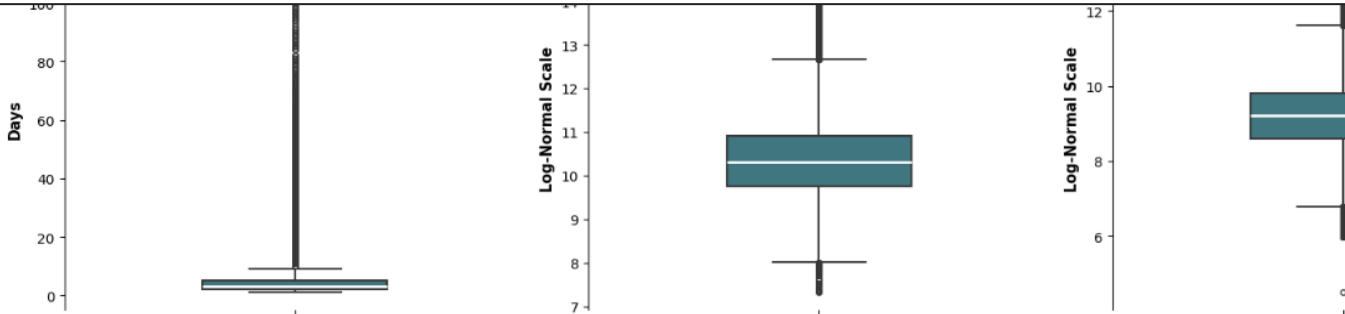
Length of Stay

01234

Total Charges1e6

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Total Costs1e6



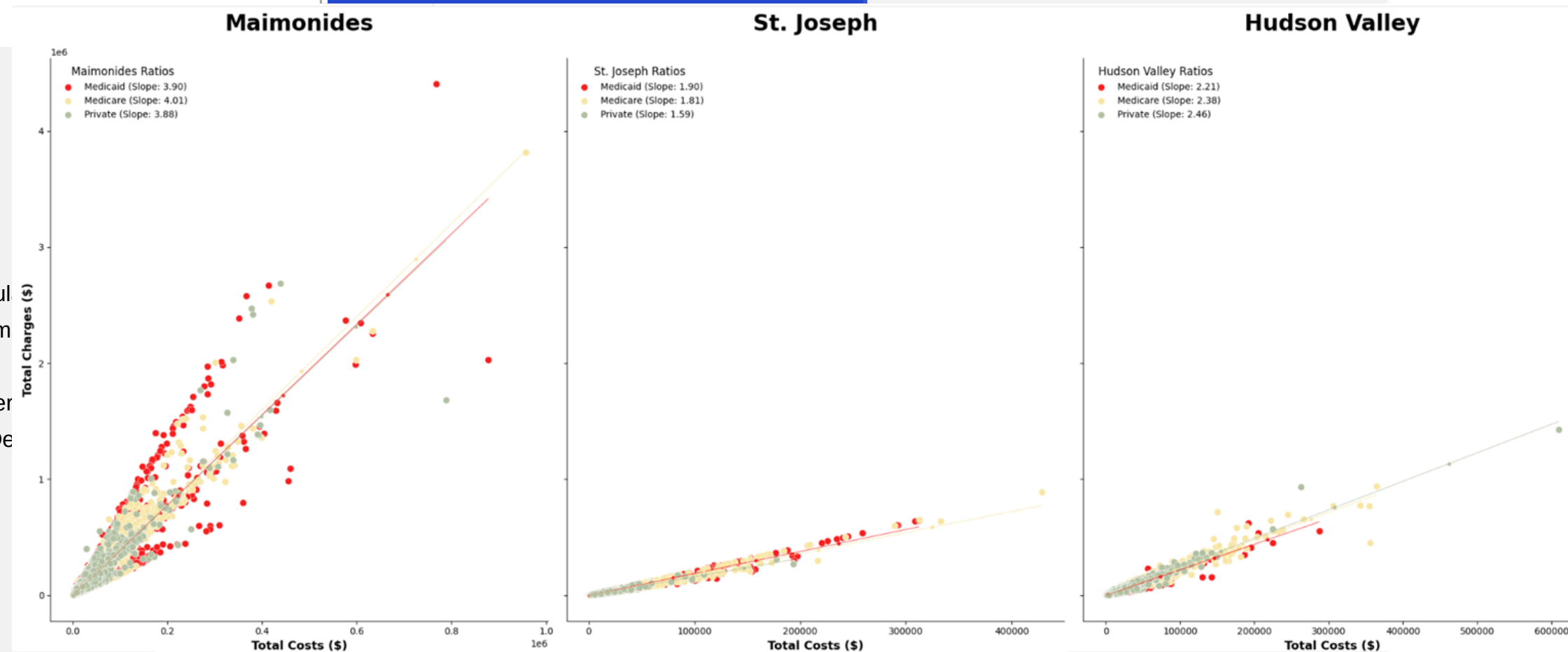
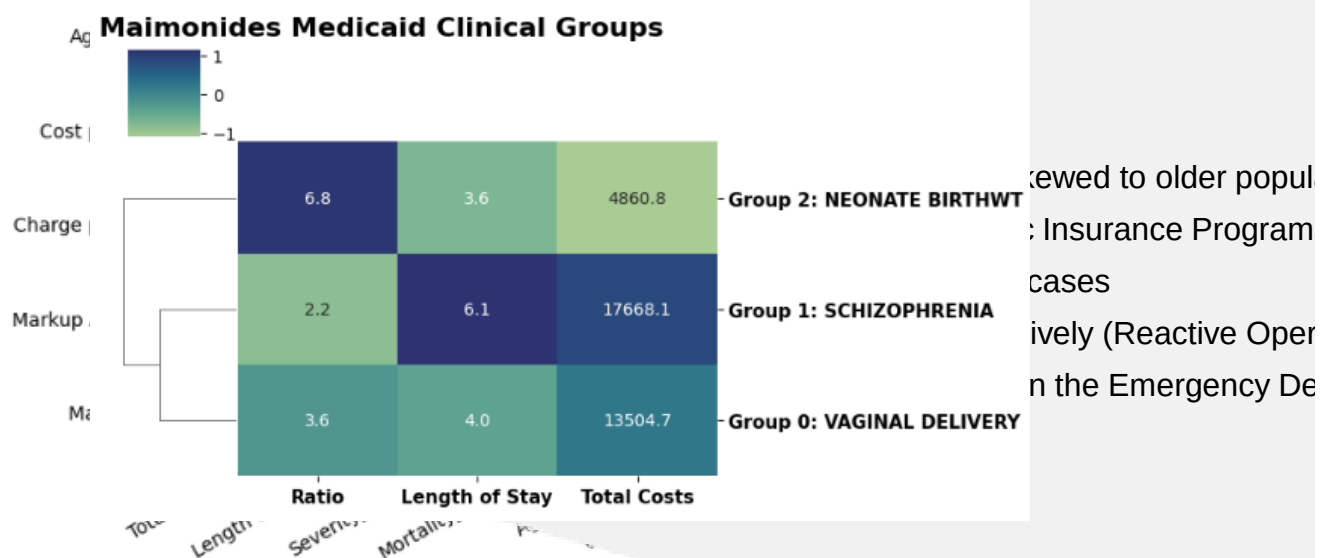
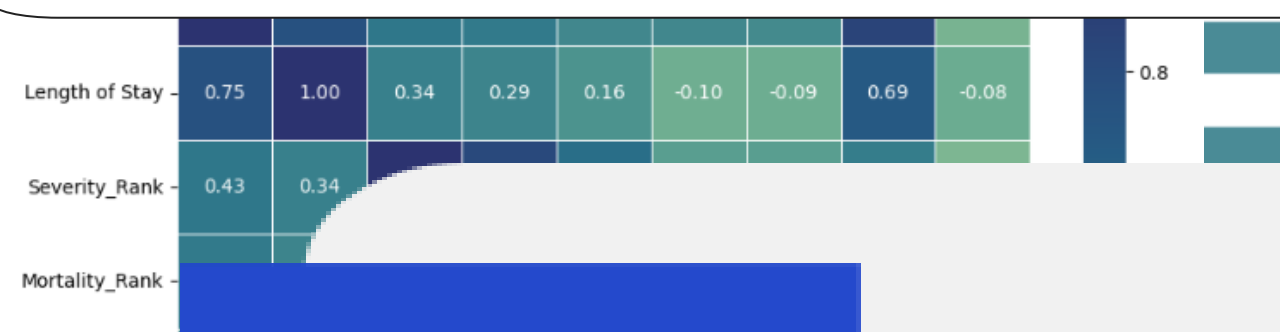
The Raw Data Analysis

Bi-variate Analysis

Categorical Feature Distribution

Bi-Variate Analysis - Correlation Matrix

Bi-Variate Analysis - Scatter Plots



Margin is determined by the Hospital contractual agreements with Insurances and the details of the Diagnosis

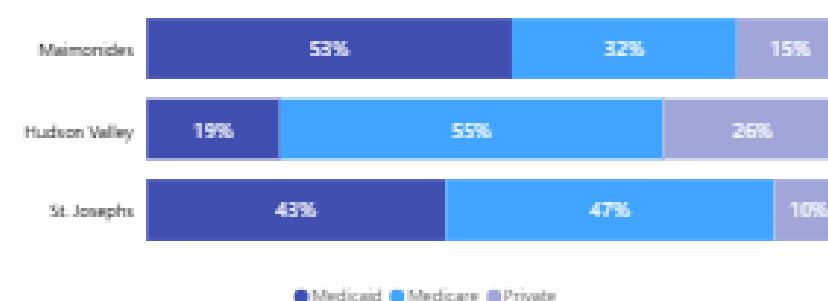
Technical Dashboard

Facility Name				Year		
Deselect All	Hudson Valley	Maimonides	St. Josephs	Deselect All	2018	2019

Avg. CCR 3,65 Charge-to-Cost	High Severity rate 30,1% Major + Extreme	ED Admission rate 53,7% Reactive Operations	Avg. Cost per Day \$3,76K CPD	CPD YoY change +15,9% Select 2019 to activate	Cross Subsidy Ratio 1,02 Cross Subsidy Ratio
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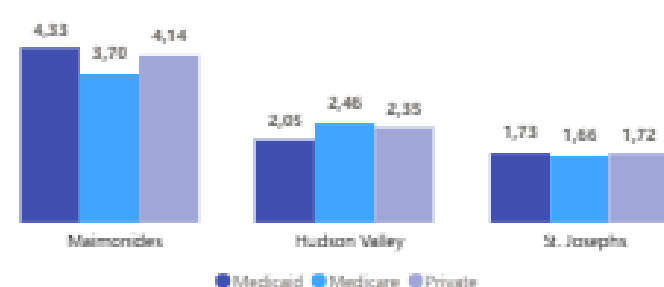
The Insurance Type

Hudson Valley is more commercialized with more cases handled by Private Health. **Maimonides** and **St. Josephs** depend on Federal and State Insurances.



Large profit margins rely on the **payer mix** of each hospital. This mix and the administrative hospital power is responsible for CCR

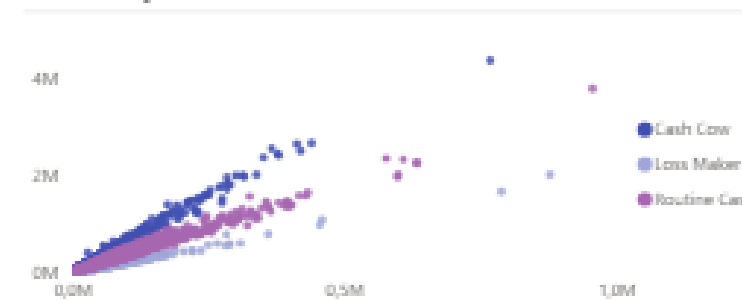
Charge-to-Cost Ratio for Hospital & Payer Type



The Diagnosis

Maimonides and Medicaid cases : Discrete triplet of charges for the same cost show strong pricing strategy, administrative and negotiation power of the hospital

Total Charges vs Total Costs



Cash Cows and Loss Makers

Neonate Birth and Schizophrenia take the lead respectively. Schizophrenia reaches a LOS of 20. Cash Cows have no representation at St. Joseph

Scatter Plot Profit Group



The Forecast

St Joseph is forecasted to have the largest Cost increase and the smallest Profit Margin Expansion with the maximum average patient stay of all three hospitals

Safety Buffer Comparison

Hospital	Cost Increase/yr	Revenue Increase /yr	Margin Expansion /yr	Avg LOS
St. Josephs	\$2.457	\$3.113	\$656	7,77
Maimonides	\$1.463	\$3.967	\$2.503	5,00
Hudson Valley	\$469	\$1.796	\$1.327	4,05

Maimonidis thrives on easy cases and through this they fund the losses they take on difficult cases (subsidization). This not reproduced in the other hospitals and it would be a case study to implement

Easy Wins & Definite Losses

Hospital	CashCow \$	Routine Case \$	Loss Maker \$	CSR
Maimonides	\$640.319K	\$2.525.859K	\$198.186K	3,23
Hudson Valley	\$770K	\$60.885K	\$304.118K	0,00
St. Josephs		\$117K	\$127.749K	

Actions to Escalate to C Board :

Increase St. Joseph's ~**1.8x markup** by adopting the Maimonides ~**4.2x Charge-to-Cost Ratio** model. Identify "Cash Cow" services to **subsidize** rising operational costs and LOS, establishing a critical "Margin of Safety.". "Cash Cows" can be identified from the graphs above.

Director Dashboard

Performance is concentrated in Maimonides and Public Payers, while smaller hospitals present inefficiencies: Strategic Decisions at Executive Level Required

Revenue

5.66bn

Costs

1.72bn

Profit

3.94bn

Profit Margin %

69.67

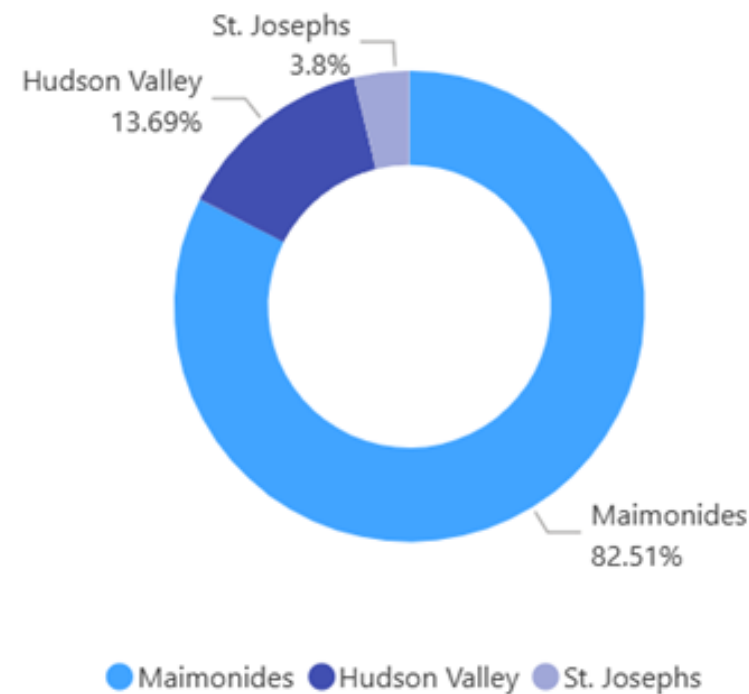
Profit to Cost %

229.73

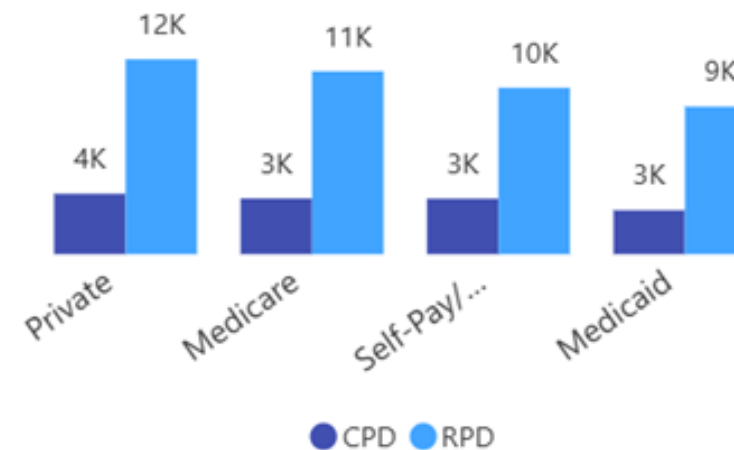
CPD

3.15K

Group health depends heavily on Maimonides

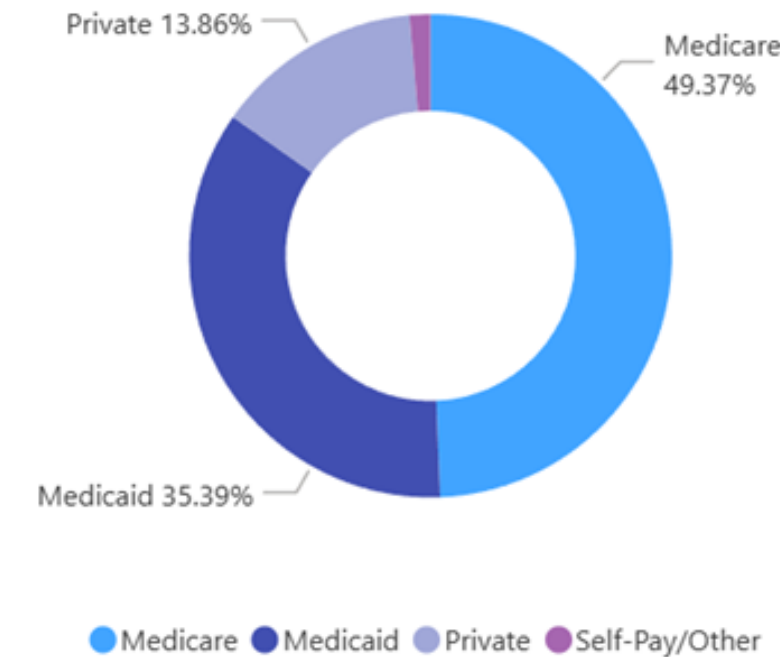


The bar chart and table demonstrate large differences in scaled KPIs across facilities, clearly highlighting operational inefficiencies.



Facility	CPD / RPD	Avg LoS
St. Josephs	0.59	7.77
Maimonides	0.27	5.00
Hudson Valley	0.43	4.05

State and federal insurance programs contribute 85% of total revenue.



Facility

- Select all
- Hudson Valley
- Maimonides
- St. Josephs

Payment

- Select all
- Medicaid
- Medicare
- Private
- Self-Pay/Other

Strategic Actions:

- Maimonides contributes 82.5% of total revenue, while federal and state insurance programs account for 85%.
- ✓ Directors should ensure C-level initiatives on state partnerships are treated as a top priority and executed consistently.
- ✓ LoS is not our top priority.

Escalation Points to the C-Level:

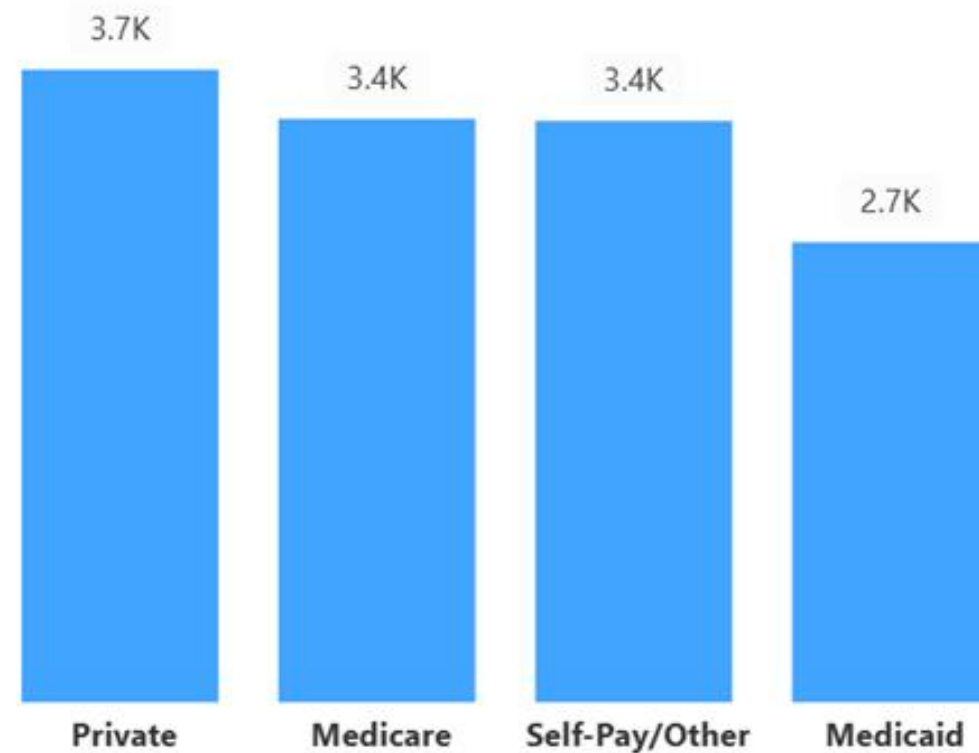
Given their limited scalability, these facilities may require strategic, top-level decisions regarding resource allocation, or restructuring. Neonate births represent a potential opportunity for St. Joseph's and Hudson Valley to improve efficiency and overall performance.

C-Level Dashboard

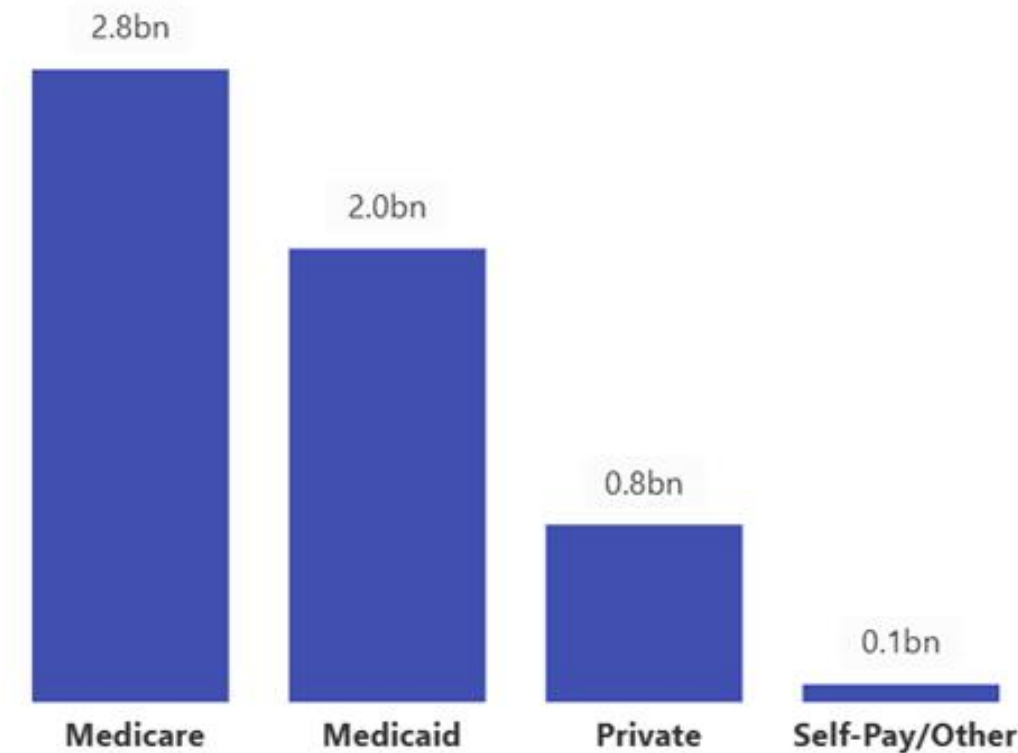
Maimonides and Public Payer Programs Drive Group Revenue



Maimonides maintains healthy costs across all payment types



State and federal programs drive high-value revenue.



Facility

- ☐ Select all
- ☐ Hudson Valley
- ☐ Maimonides
- ☐ St. Josephs

Payment

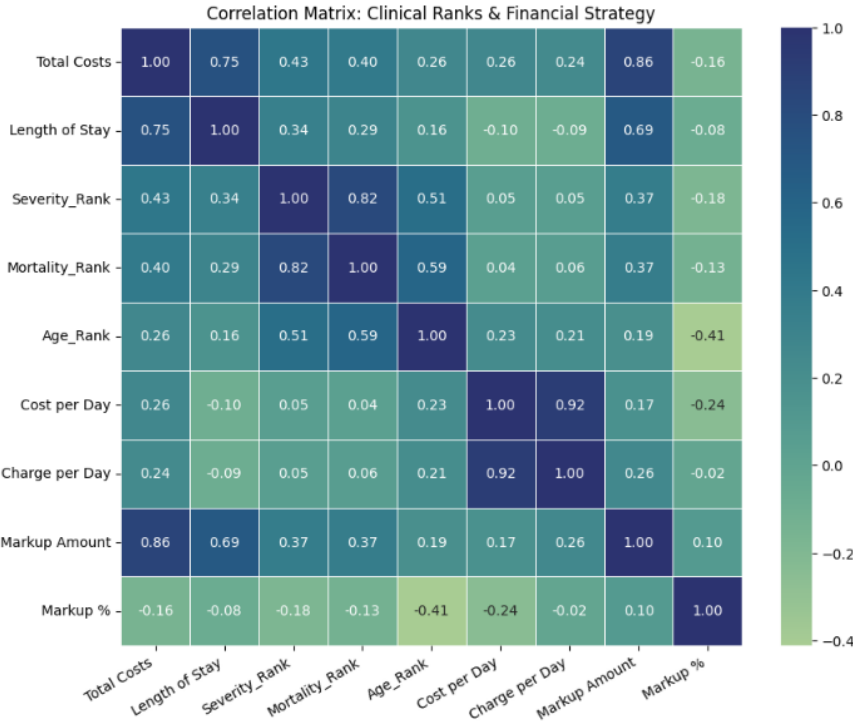
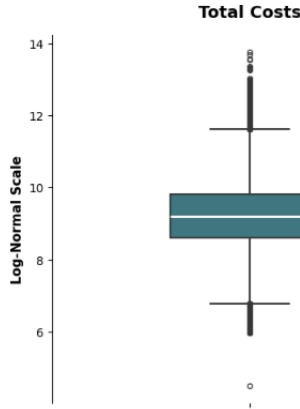
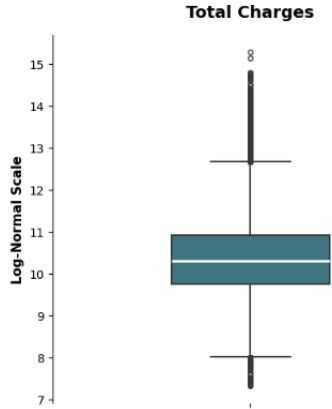
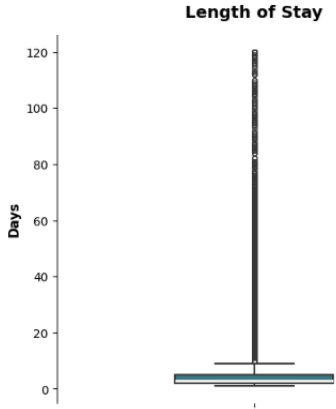
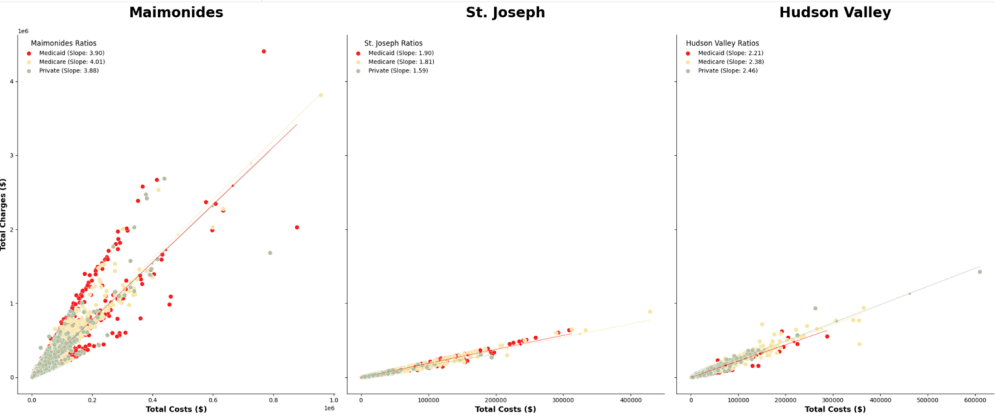
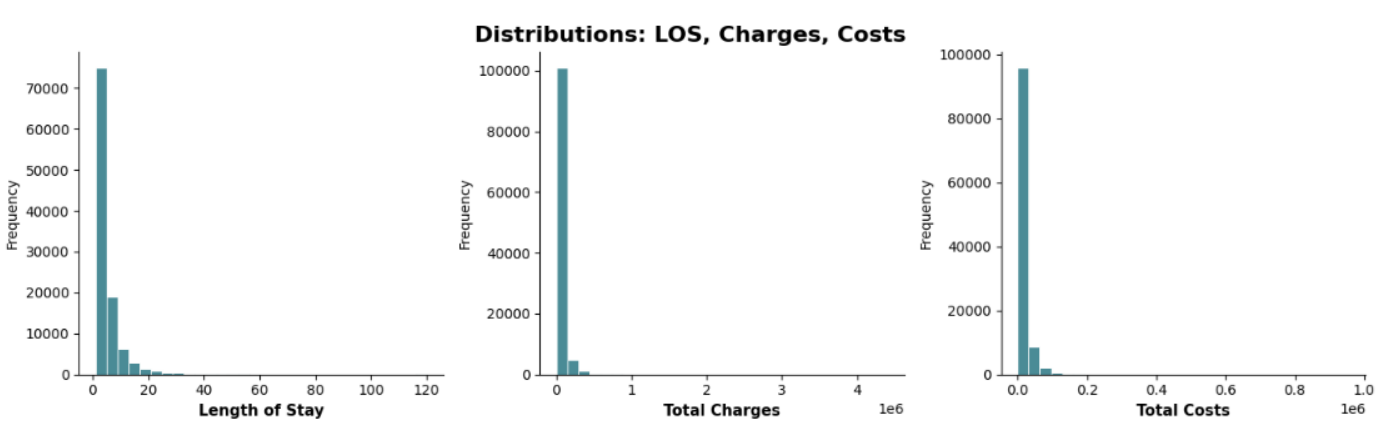
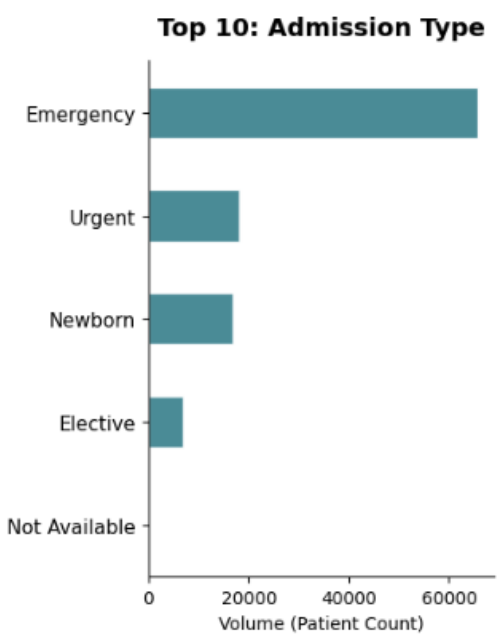
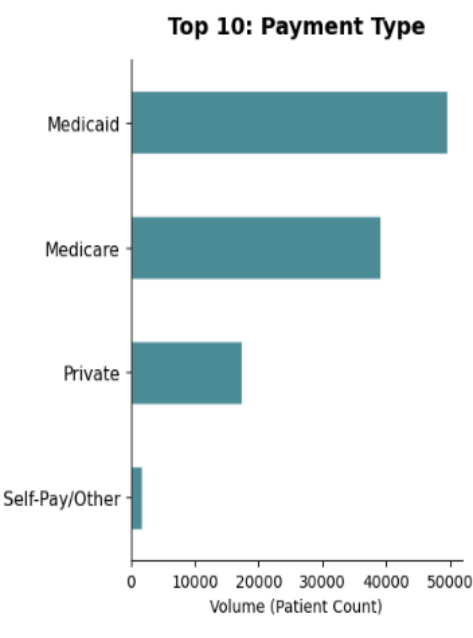
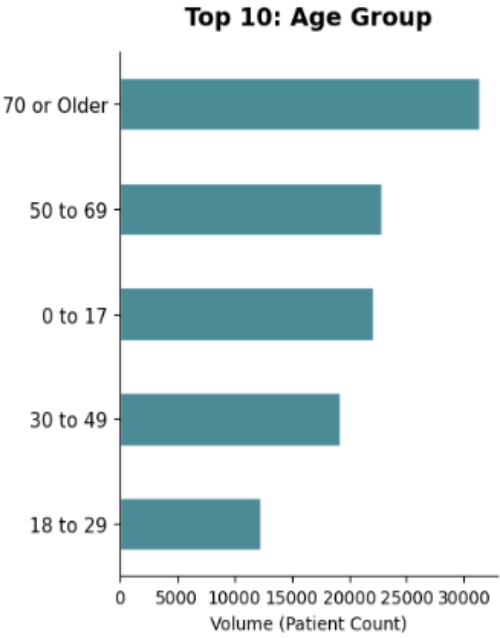
- ☐ Select all
- ☐ Medicaid
- ☐ Medicare
- ☐ Private
- ☐ Self-Pay/Other

Strategic Actions:

Maintaining strong relationships with federal and state payers is essential to protect and stabilize high-revenue Maimonides flows. The group should consider targeted (re)investment in Maimonides to further strengthen efficiency and revenue outcomes. Bold decisions are required for St. Joseph's: A possible strategic move of staff to Maimonides could reduce overall operational costs while addressing Maimonides' potential understaffing, especially since scaling up St. Joseph's is not operationally feasible. St. Joseph's could be restructured and specialized in normal neonate birth management, which directly enhances its financial position.



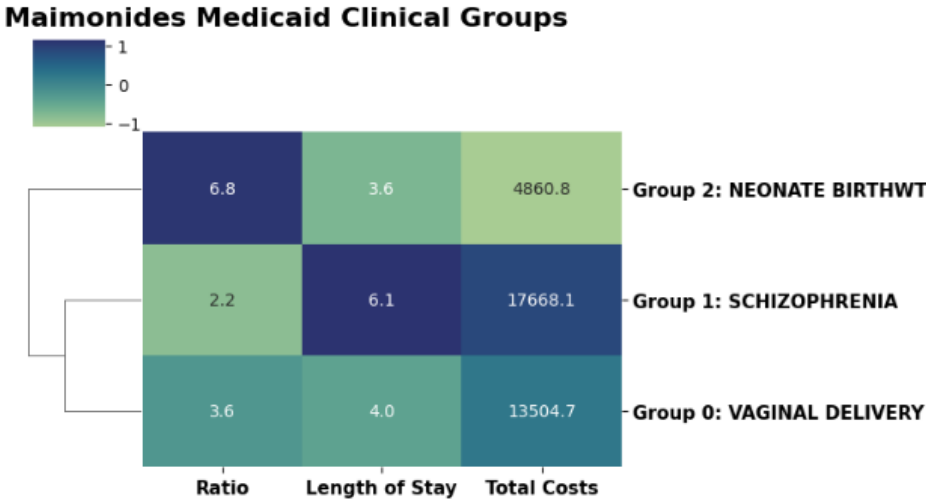
THANK YOU!



```
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summary["skew"] = df[numeric_cols].skew(numeric_only=True)
summary
```

	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max	skew
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Total Costs	107682.0	15946.164155	22726.076002	88.27	791.1400	947.5245	5410.03	9974.455	18189.6025	49996.698	103110.3964	956351.23	8.046440

Appendix



- Maimonides dominates
- Most people are of >70 years of age (skewed to older population)
- Emergency Admission dominates massively (Reactive Operations)
- 80% of emergency admissions end up in the Emergency Department
- Most payments are by Medicaid (public Insurance Programs)
- Most admissions are for minor severity cases

- **0.75 Correlation between (LOS) and Total Costs.** This proves that the hospital's primary cost driver is "time in bed". Any policy involving LOS will have a big impact on bottomline
- **-0.10 LOS and Cost per Day** . The expensive health actions happens in the early days.
- **0.86 Total Cost and Markup amount.**
- -0.16 Total Cost and Markup % . When Costs are high but profit margin is efficiency is small .
- **0.19 of Age_Rank and Markup Amount.** As age increases in brings slightly more dollars
- **-0.41 Age_Rank with Markup %.** As patients get older, the hospital's profit buffer becomes smaller
- **0.82 High severity and high mortality**

Appendix